CSEDU Digital Knowledge Platform

Software Requirements Specification

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Version:	2
Status:	Draft
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Department of Computer Science and Engineering, University of Dhaka

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1 Document Control

1.1 Metadata

Field	Value
Team	Team Devops
Project Name	CSEDU Digital Knowledge Platform with AI Agent
Repository URL	Repository Link
Version	2
Status	Draft
Date	09/04/2026

Table 1: Document Metadata

1.2 Authors and Reviewers

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1.3 Revision History

Version	Date	Author	Changes
2	09/04/2026	Team Devops	Revised and modified version

Table 2: Revision History

2 Executive Summary

The CSEDU Digital Knowledge Platform is an integrated system designed to consolidate academic resources, research outputs, student projects, and library operations. This version of the Software Requirements Specification introduces a sophisticated **Retrieval-Augmented Generation (RAG)-based AI Agent** that acts as an intelligent assistant across all platform modules.

The RAG AI Agent serves as a unified knowledge system capable of:

- Summarizing research papers and academic documents automatically
- Answering user queries about research papers, student projects, and library resources
- Providing intelligent insights and recommendations based on departmental archives
- Enabling semantic search across heterogeneous content types
- Supporting both English and Bangla natural language queries

This enhancement significantly improves platform usability, discoverability, and user engagement while maintaining all core functionalities of the digital archive, research repository, student showcase, and library management system.

3 Problem Definition

3.1 Problem Statement

What problem are you solving?

CSEDU currently lacks a unified system to:

- Manage academic resources, research outputs, student projects, and library operations in one place
- Enable intelligent discovery and semantic understanding of academic content
- Provide users with contextual insights about departmental knowledge and resources
- Leverage modern AI technologies to improve accessibility and reduce information discovery overhead

Resources are scattered across multiple platforms, library workflows require manual processes, and users struggle to discover relevant academic materials without significant search effort. Additionally, there is no intelligent system to answer domain-specific questions or provide contextual understanding of the institution's collective knowledge.

Why does it matter?

A unified platform with integrated AI capabilities will:

- Streamline storage, discovery, and access to institutional knowledge
- Reduce administrative overhead for librarians and faculty
- Vastly improve visibility and accessibility of research and academic materials
- Enable students, faculty, and public to discover and understand resources through natural language interaction
- Facilitate knowledge transfer and insight discovery across departmental archives
- Support multi-language access for broader accessibility (English and Bangla)

3.2 Target Users

User Group	Description	Main Needs
Students	Undergrad/MsC/PhD students	Search and download library catalog, papers and academic resources, borrow books, showcase hobby and departmental projects, track book borrowing history, ask AI agent about research topics and student projects.
Researchers	Faculty and Researchers	Upload and download research, manage media metadata, version drafts, manage and access restricted archives and research papers, receive AI-generated summaries and insights.
Librarians	Library administration staff	Manage lending/returns of books and student membership, track overdues, bulk upload catalog data, scan barcodes, use AI agent to answer patron queries about catalog.
Public Users	Non-registered visitors	Browse public archives, view the student project showcase, download publicly available research papers, search the library catalog using AI-powered discovery.
Administrators	System managers	Manage platform and permissions, configure AI Agent models and parameters, monitor AI performance and usage patterns.

Table 3: Target User Groups

3.3 Goals and Non-Goals

Goals:

- Create a centralized digital archive
- Unify four core components into one web platform with integrated AI capabilities
- Enable research publication and sharing with AI-assisted discovery
- Showcase student projects with intelligent curation and recommendation
- Provide a searchable digital library with semantic understanding
- Support multi-format media with AI content analysis capabilities
- Digitize library workflows with AI-assisted patron support
- Implement robust access tiers with context-aware AI responses
- **Deploy a RAG-based AI Agent for intelligent knowledge discovery and synthesis**
- **Enable natural language querying across all platform modules**
- **Provide automated summarization of research papers and documents**

Non-goals:

- General university HR management
- Grading/LMS integration (e.g., Moodle/Canvas replacement)
- Financial accounting beyond library fines
- Not designed for commercial book sales
- Not designed to replace peer-review systems (AI summaries are supplementary)
- No advanced personalized AI recommendation engine (beyond basic semantic similarity)
- No real-time collaborative editing
- No external payment gateway integration

4 Scope and Context

4.1 Project Overview

The CSEDU Digital Knowledge Platform consolidates four tightly integrated modules into a single, cohesive web-based system, enhanced with a sophisticated **RAG-based AI Agent** that serves as a central intelligence layer:

1. **Digital Archive** — A centralised repository for storing, organising, and retrieving multi-format academic and institutional media with rich metadata and lifecycle management.
2. **Research Repository** — A structured portal for faculty and researchers to publish, version, and share research outputs, lab portfolios, and working papers, with AI-assisted discovery and summarization.
3. **Student Project Showcase** — A browsable, publicly accessible gallery of student projects with AI-powered curation, filtering, and recommendation capabilities.
4. **Digital Library & Catalog** — A fully digitised library management system supporting physical lending workflows, catalog search, member account management, and librarian operations, enhanced with AI-assisted patron support.
5. **RAG-Based AI Agent** — An intelligent assistant that understands platform content, answers user queries, provides summarization, delivers insights, and enables semantic discovery across all modules.

4.2 In Scope

4.2.1 Digital Archive

- Upload, storage and download of multi-format media including PDF, DOCX, audio, video, and images
- Rich metadata entry per item: title, abstract, authors, keywords, language, date, format, and custom tags
- Bulk upload of media and metadata via structured CSV/spreadsheet import
- In-place metadata editing by authorised staff and researchers
- Document lifecycle management with four distinct states: **Draft** → **Under Review** → **Published** → **Archived**
- Version control: each edit to a published item creates a new numbered version while retaining all prior versions

- Tiered access control governing visibility at four levels: **Public**, **Member**, **Staff**, and **Restricted**
- Integrated media viewer supporting in-browser PDF rendering and audio/video playback
- **Automatic content extraction and indexing for AI Agent knowledge base**
- **AI-powered semantic tagging and content classification**

4.2.2 Research Repository

- Dedicated submission portal for research papers, conference proceedings, technical reports, and lab portfolios
- Co-author tagging and ORCID-style contributor attribution fields
- Draft/review/publish workflow with reviewer assignment (internal staff only)
- Public-facing research pages listing outputs by author, lab, or topic
- Restricted pre-publication drafts visible only to the submitting researcher and designated reviewers
- **Automatic paper summarization and abstract enhancement via AI Agent**
- **AI-generated research keywords and topic detection**
- **Intelligent author collaboration suggestions based on research similarity**
- **AI-powered citation and reference analysis**

4.2.3 Student Project Showcase

- Public gallery displaying completed student projects as browsable cards/tiles
- Filter and sort controls by academic year, degree level (BSc/MSc/PhD), and topic/tag
- Per-project detail page showing abstract, team members, supervisor, and attached media
- Student self-submission workflow (subject to staff approval before publication)
- **AI-assisted project discovery based on student interests and skill areas**
- **Automatic project summarization and topic extraction**
- **AI-powered relevance ranking and intelligent filtering**

4.2.4 Digital Library & Catalog

- Comprehensive library catalog with advanced search supporting title, author, ISBN, subject, and keyword queries in both **English** and **Bangla**
- Physical item management: add, edit, deactivate, and delete catalog records via full Librarian CRUD operations
- Barcode and QR code support for rapid identification of physical items during checkout and return operations
- Lending workflow: checkout, return, renewal, and due-date tracking per member
- Overdue detection with automatic fine calculation based on configurable per-day rate rules
- Fine management: view outstanding fines, record payments, and enforce checkout blocks for members exceeding fine limits
- Hold/reservation system: members may place holds on checked-out items and receive notification upon availability
- Member account features: active loan view, borrowing history, wishlist, hold requests, and fine balance dashboard
- Librarian operations dashboard: real-time item status, overdue alerts, and one-click checkout/return via barcode scanner input
- Reporting module: overdue item reports, fine collection summaries, and catalog usage statistics
- CSV/bulk import tool for migrating legacy catalog records into the system
- **AI Agent-powered patron support chatbot for catalog queries**
- **Intelligent book recommendation engine based on patron interests and borrowing history**
- **Natural language catalog search with semantic understanding**
- **AI-assisted overdue notification with personalized messaging**

4.2.5 RAG-Based AI Agent System

- **Architecture:** Retrieval-Augmented Generation system leveraging vector embeddings and large language models
- **Content Ingestion:** Automatic extraction and processing of text from PDFs, documents, research papers, project descriptions, and catalog metadata
- **Vector Database:** Maintenance of semantic embeddings for all platform content

enabling similarity-based retrieval

- **Query Processing:** Support for natural language questions in English and Bangla with automatic language detection
- **Answer Generation:** RAG-based synthesis combining retrieved relevant content with generative AI for contextual responses
- **Document Summarization:** Automatic generation of summaries for research papers, student projects, and long documents
- **Insight Generation:** Extraction of key findings, trends, and patterns from research outputs and institutional knowledge
- **Context Awareness:** Access control integration ensuring AI responses respect user permission tiers
- **Conversation History:** Maintenance of user chat sessions with context for multi-turn conversations
- **Performance Metrics:** Tracking and logging of AI Agent usage patterns, query types, and response quality
- **Explainability:** Citations of source documents in AI-generated responses for transparency and verifiability
- **Feedback Loop:** User feedback mechanism to continuously improve AI response quality
- **Multilingual Support:** Full support for English and Bangla with seamless code-switching

4.2.6 User Account Management

- Self-registration and login for students and staff using institutional credentials
- Role assignment and tier management by administrators (Public → Member → Staff → Admin)
- Profile management: name, contact details, and notification preferences
- Activity log visible to the account holder (uploads, loans, holds, fines)
- **AI Interaction Preferences:** User settings for AI summary length, detail level, and preferred languages
- **Personalized Dashboard:** AI-curated recommendations and insights based on user profile and history

4.2.7 Librarian Management Dashboard

- Unified dashboard consolidating all librarian-facing operations
- Barcode/QR scanner input field for rapid checkout and return processing
- Item lookup by barcode, title, or member ID
- Overdue and fine management panels with exportable reports
- Bulk metadata import and catalog entry management tools
- **AI Assistant Chat Widget:** Quick access to AI Agent for answering patron queries about catalog and availability
- **AI-Generated Reports:** Automated insights on circulation patterns, popular items, and collection gaps

4.3 Out of Scope

The following are explicitly excluded from the current project iteration:

- Direct integration with university tuition fee or general payment gateways (library fine payments are tracked internally but not processed via external payment APIs)
- Automated plagiarism checking or similarity detection for uploaded research or projects (AI Agent provides summaries only)
- Integration with external academic databases (e.g., Scopus, IEEE Xplore, Google Scholar API)
- Real-time chat, messaging, or forum communication between users (AI Agent chat is unidirectional)
- General-purpose personalized AI recommendation engine (beyond semantic similarity)
- Integration with the university's existing LMS platforms (e.g., Moodle, Canvas)
- General university HR management or payroll functions
- Financial accounting beyond the tracking of library fines
- Real-time content moderation or automated censorship (flagging only)
- Training of custom AI models from scratch (utilization of pre-trained LLMs)

4.4 Key Deliverables

Deliverable	Description
Archive Upload & Metadata Editor	Staff-facing interface for uploading multi-format files, entering metadata, and managing document lifecycle states and versions.
Public Search, Browse & Media Viewer	Publicly accessible catalog and archive search with bilingual support, faceted filters, and an integrated in-browser media viewer.
Research & Showcase Pages	Public-facing portals listing research outputs by author/lab and displaying the student project showcase gallery with filtering controls.
Public Catalog Portal	Searchable library catalog accessible to unauthenticated users showing item availability, location, and basic metadata.
Member Borrowing Workflow	Authenticated member interface for placing holds, tracking active loans, viewing borrowing history, managing wishlists, and checking fine balances.
Librarian Operations Dashboard & Reports	Comprehensive librarian dashboard for barcode-driven checkout/return, overdue management, fine tracking, bulk imports, and exportable reports.
RAG-Based AI Agent Chat Interface	User-facing chatbot interface enabling natural language queries across all platform content with context-aware responses and source citations.
AI-Powered Summarization Engine	Automated system for generating summaries of research papers, student projects, and documents with quality control.
AI Agent Administration Panel	Backend interface for managing AI models, vector database, query logs, and performance metrics.

Table 4: Key Project Deliverables

4.5 Assumptions and Constraints

- **Technical – File Size:** The system supports file uploads with defined limits: Video $\leq$ 200MB, PDF $\leq$ 100MB, Image $\leq$ 10MB, Audio $\leq$ 50MB. These limits must be enforced at upload time.
- **Technical – Upload Performance:** Files $\leq$ 20MB should upload within 5 seconds in at least 90% of cases under normal network conditions. Upload progress feedback must be displayed to users.
- **Technical – Bilingual Support:** All system components including UI labels, metadata, search indexing, and AI responses must support UTF-8 encoded English and Bangla text.
- **Technical – Authentication and SSO:** The system assumes integration with institutional identity providers using OAuth 2.0 or SAML. Session management must follow defined timeout and token policies.
- **Technical – Barcode Hardware:** Librarian workstations will use HID-compatible barcode/QR scanners; no proprietary drivers or SDK integration is required.
- **Technical – AI Infrastructure:** The system uses pre-trained language models combined with vector-based retrieval for semantic search and response generation. Deployment may be API-based or self-hosted.
- **Technical – AI Response Behavior:** AI responses are expected within 2–10 seconds depending on query complexity. The system must provide loading indicators and handle delayed responses gracefully.
- **Technical – AI Access Control:** AI components must enforce the same role-based access restrictions as the core system. Restricted documents must not be exposed in responses.
- **Technical – AI Output Reliability:** AI-generated outputs are supplementary and must include source references where applicable. Low-confidence responses should be avoided or flagged.
- **Technical – Data Integrity:** All transactions (e.g., borrowing, payments, uploads) must maintain consistency. Failed operations must not result in partial or corrupted data.
- **Organisational – Data Migration:** Existing catalog and archive data must be migrated prior to deployment. All required metadata fields must be validated during migration.
- **Organisational – Fine and Payment Rules:** Fine calculation rules (rate, thresholds) will be defined by the library administration. All payments must be linked to

valid fine records.

- **Organisational – Payment Processing:** The system assumes integration with an external payment gateway. Payment status (Pending, Successful, Failed) is determined by the gateway response.
- **Organisational – External Users:** External researchers may access only publicly available or explicitly permitted content. Restricted materials require institutional authentication.
- **Organisational – Content Review Workflow:** Uploaded research content must undergo a review and approval process before becoming publicly accessible.
- **Timeline:** The MVP must include catalog search, lending workflow, fine management, SSO authentication, and basic AI query functionality. Advanced AI features and research portals are part of the full release.

5 Stakeholders and Roles

5.1 Stakeholders

Stakeholder	Interest	Priority
CSEDU Dept Head	Overall system success, academic visibility, AI-enhanced knowledge discovery	High
Librarian	Efficiency of cataloging and circulation, AI-assisted patron support	High
Students	Ease of access to materials and showcase, AI-powered discovery and learning aids	High
Faculty	Ease of access to material and publishing research outputs, AI summarization	High
Researchers	Publishing research outputs, AI-enhanced discoverability	Medium
IT/DevOps	System stability, AI infrastructure management, performance optimization	High

Table 5: Stakeholder Register

5.2 User Roles and Permissions

Role	Description	Permissions
Public	Unauthenticated user	View public catalog, showcase, and public archives. Access AI Agent for general queries on public content.
Member	Authenticated student/staff	Place holds, view history, access member-tier archives. Full AI Agent access with personalized recommendations.
Faculty / Researcher	Contributors	Upload/edit media, manage versions (draft/review/publish). Access AI summarization tools. Receive AI-suggested improvements.
Librarian	Library manager	Full CRUD on library items, manage lending, manage fines. Access AI Agent for patron support. View AI-generated circulation reports.
Admin	System administrator	Manage user roles, access tiers, and system configurations. Administer AI Agent settings and models. Monitor AI performance.
AI Administrator	AI system specialist	Configure AI models, manage vector database, optimize queries, monitor performance metrics, adjust AI parameters.

Table 6: User Roles and Permissions

6 Functional Requirements

Requirement ID format: `FR-TXX-###` *(Core Platform),* `FR-AI-###` *(AI Agent)*

6.1 Core Platform Requirements

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-TXX-001	Media Upload	The system shall allow staff to upload multi-format media (PDF, video, image) with metadata.	M	• Supported formats: PDF, MP4, JPG, PNG • File size limits: PDF $\leq$ 100MB, Image $\leq$ 10MB, Video $\leq$ 200MB • Upload completes within $\leq$ 5 seconds for files $\leq$ 20MB (90% cases) • Metadata fields (title, author, tags) are stored and searchable

FR-TXX-002	Bilingual Search	The system shall support catalog search using both English and Bangla queries.	M	• Supports UTF-8 input for Bangla and English • Search results returned within $\leq$ 2 seconds (90% cases) • Returns at least one relevant result if data exists • Language-specific tokenization applied
FR-TXX-003	Circulation	The system shall track lending and return using Barcode/QR.	M	• Barcode scan updates item status • Borrowed / Returned states correctly reflected • Transactions stored in circulation log
FR-TXX-004	Tiered Access	The system shall restrict document access based on role.	M	• Unauthorized access returns HTTP 403 • Role permissions enforced for all documents

FR-TXX-005	Fine Tracking	The system shall calculate overdue fines automatically.	S	• Fine calculated when item becomes overdue • Daily rate configurable (default: 50 BDT/day)
FR-TXX-006	User Registration	System shall allow users to create accounts.	M	• Email validation required • Password $\geq$ 8 characters • Duplicate accounts prevented
FR-TXX-007	Login	Users shall login using credentials.	M	• Valid login grants access • Invalid login shows error • Account locked after 5 failed attempts
FR-TXX-008	Project Showcase	Students shall publish projects.	M	• Project page created and accessible • Metadata visible to users
FR-TXX-009	Research Repository	Researchers shall upload research outputs.	M	• Research entries stored with metadata • Entries searchable

FR-TXX-010	Library Catalog	Users shall browse catalog.	M	• Results paginated (10 per page) • Includes title, author, availability • Response time $\leq$ 2 seconds
FR-TXX-011	Borrow Book	Members shall borrow books.	M	• Status updated to "Borrowed" • Due date assigned • Transaction recorded
FR-TXX-012	Return Book	Librarians shall record returns.	M	• Status updated to "Available" • Fine calculated if overdue • Transaction updated
FR-TXX-013	Hold Request	Members shall reserve books.	S	• Hold request stored • User notified when available
FR-TXX-014	Fine Management	System shall manage overdue fines.	C	• Fine = (Current Date - Due Date) $\times$ Rate • Fine visible in dashboard • Updates every 24 hours
FR-TXX-015	Content Versioning	System shall maintain version history.	M	• All edits tracked • Previous versions retrievable

FR-TXX-016	Metadata Bulk Import	System shall allow CSV bulk upload.	S	• CSV parsed into multiple records • Validation errors reported
FR-TXX-017	Fine Payment	System shall allow users to pay fines.	M	• User views fine breakdown • Payment recorded and confirmed • Status updated to "Paid" • Borrowing restriction removed
FR-TXX-018	Checkout Restriction	System shall block borrowing for unpaid fines.	M	• Block if fine > 50 BDT • Error message displayed • Allowed after payment
FR-TXX-019	External Researcher Access	System shall support external researchers.	S	• Access to public content only • No borrowing allowed • Restricted content returns 403
FR-TXX-020	Single Sign-On (SSO)	System shall support institutional login.	M	• OAuth 2.0 / SAML supported • Token expiry ≤ 1 hour • Max 5 failed attempts per minute • User account mapped/created

6.2 RAG-Based AI Agent Requirements

FR-AI-001	AI Chat Interface	The system shall provide a conversational chat interface for querying platform content.	M	• Chat interface accessible via web UI • User messages processed and responses returned • Response time $\leq$ 5 seconds (90% cases)
FR-AI-002	Natural Language Processing	The AI Agent shall process natural language queries in English and Bangla.	M	• Supports UTF-8 input for Bangla and English • Returns contextually relevant responses • Handles mixed-language queries
FR-AI-003	Document Summarization	The AI Agent shall generate summaries of documents.	S	• Summary generated within $\leq$ 10 seconds (90% cases) • Output length: 100–300 words (configurable) • Includes at least 3 key points

(AI Agent Requirements – continued)

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-AI-004	Semantic Search	The AI Agent shall perform semantic search using embeddings.	M	• Returns relevant results despite keyword variation • Results ranked by relevance score • Query response time ≤ 3 seconds (90% cases)
FR-AI-005	Answer Generation with Citations	The AI Agent shall generate answers with source references.	M	• Responses include document references • Each reference includes clickable link • No hallucinated sources allowed
FR-AI-006	Bilingual Responses	The AI Agent shall respond in the user's query language.	M	• Automatic language detection • Output matches input language (English/Bangla)
FR-AI-007	Access Control Integration	The AI Agent shall enforce user permission constraints.	M	• Only accessible content used in responses • Restricted content never exposed

(AI Agent Requirements – continued)

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-AI-008	Conversation History	The AI Agent shall maintain session context.	S	• Context retained for at least 10 messages • Multi-turn responses remain coherent
FR-AI-009	Research Insights	The AI Agent shall extract structured insights from research papers.	S	• Includes: – Key Findings ($\geq$ 3 bullet points) – Methodology summary – Conclusion summary • References source document
FR-AI-010	Project Intelligence	The AI Agent shall extract structured information from student projects.	S	• Includes: – Technologies used – Skills demonstrated – Project outcome

(AI Agent Requirements – continued)

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-AI-011	Catalog Assistance	The AI Agent shall answer queries about library resources.	M	• Provides availability status • Suggests related resources • Response time $\leq$ 3 seconds
FR-AI-012	Content Classification	The AI Agent shall classify and tag content automatically.	S	• Assigns subject tags • Classification accuracy $\geq$ 80%
FR-AI-013	Intent Recognition	The AI Agent shall detect user intent.	M	• Supported intents: – Question answering – Summarization – Recommendation • Accuracy $\geq$ 80%
FR-AI-014	Vector Database Management	The system shall maintain embeddings for all content.	M	• New content indexed automatically • Embeddings searchable within $\leq$ 5 seconds

(AI Agent Requirements – continued)

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-AI-015	AI Performance Monitoring	The system shall track AI performance metrics.	S	• Logs response time, usage count • Admin dashboard displays metrics
FR-AI-016	User Feedback	The AI Agent shall collect feedback on responses.	S	• Users can rate responses • Feedback stored for analysis
FR-AI-017	Recommendations	The AI Agent shall provide personalized suggestions.	C	• Recommendations based on user activity • At least 3 suggestions per query
FR-AI-018	Restricted Query Handling	The AI Agent shall block unauthorized queries.	M	• Returns "Access Denied" for restricted queries • No restricted data exposed
FR-AI-019	AI Failure Handling	The system shall handle AI failures gracefully.	S	• Fallback message shown to user • Errors logged for monitoring

Priority Legend: ***M*** *= Must have,* ***S*** *= Should have,* ***C*** *= Could have,* ***W*** *= Won't have (this iteration)*

7 User Stories and Use Cases

7.1 User Stories

US-TXX-001	Public User	to search the digital library catalog	I can see if a book is available.	Can search by title/author; sees availability status.
US-TXX-002	Student	to view the project showcase	I can see past CSEDU projects for inspiration.	Projects displayed in a grid with filters for year/topic.
US-TXX-003	Member	to place a hold request on a checked-out book	I can reserve it when it returns.	Hold is recorded; user is notified when available.
US-TXX-004	Member	to view my borrowing history and current fines	I can manage my library account.	Dashboard shows active loans, past loans, and fine total.
US-TXX-005	Researcher	to upload a draft of my paper	I can securely store it before review.	Document is saved with "Draft" status; not visible to public.
US-TXX-006	Researcher	to add rich metadata to my uploads	My work is easily discoverable.	Can tag with keywords, abstract, and co-authors.
US-TXX-007	Staff	to bulk upload metadata using a CSV	I don't have to enter hundreds of records manually.	System parses CSV and creates library catalog entries.
US-TXX-008	Librarian	to scan a book's barcode	I can quickly process a check-out/return.	Scanner input instantly brings up the item record.
US-TXX-009	Librarian	to view an overdue report	I can contact members with late items.	Generates a list of all overdue items and associated users.
US-TXX-010	Researcher	to edit research metadata	to maintain accuracy.	Metadata updated.

(User Stories – continued)

Story ID	As a...	I want...	So that...	Acceptance Criteria
US-TXX-011	Admin	to assign access tiers to users	I can control who sees restricted materials.	Admin dashboard allows changing user role from Member to Staff.
US-TXX-012	Student	to switch the UI between Bangla and English	I can use the system in my preferred language.	Language toggle translates all static UI elements.
US-TXX-013	Student	to upload my project	I can showcase my work.	Project visible after staff approval.
US-TXX-014	Member	to borrow a book	I can read resources offline.	Borrow recorded; book unavailable to others.
US-TXX-015	Member	to return a book	I can avoid accumulating fines.	Status updated; loan ends.
US-TXX-016	User	to view media online	I can avoid downloading large files.	Media viewer loads and plays in-browser.
US-AI-001	Student	to ask the AI Agent about a research paper	I can understand its main contributions without reading it all.	AI provides a summary with key points.
US-AI-002	Researcher	to get automatic summaries of papers similar to my research	I can stay informed about related work.	AI suggests and summarizes related papers.
US-AI-003	Member	to ask the AI Agent if a book is available	I can quickly get an answer without browsing the catalog.	AI responds with availability status and location.

(User Stories – continued)

Story ID	As a...	I want...	So that...	Acceptance Criteria
US-AI-004	Public User	to search for papers on a specific topic using natural language	I can find relevant research without technical keywords.	Semantic search returns papers on requested topic.
US-AI-005	Student	to get personalized project recommendations	I can find inspiration from related student work.	AI suggests projects matching student interests.
US-AI-006	Librarian	to use the AI Assistant to answer patron queries	I can reduce time spent on routine questions.	AI responds to common catalog and availability questions.
US-AI-007	Admin	to monitor AI Agent performance and usage statistics	I can optimize AI configuration and identify issues.	Admin dashboard shows query logs, response times, and user feedback.
US-AI-008	Researcher	to get AI-generated keywords and classifications for my paper	My work is properly categorized for discoverability.	AI suggests subject tags and classification categories.
US-AI-009	Faculty	to ask the AI about student projects in the showcase	I can identify potential collaborators or funding opportunities.	AI provides structured summaries of student work with skill tags.
US-AI-010	Student	to ask the AI about departmental archives	I can find historical resources and learn from previous batches.	AI retrieves and summarizes relevant archival materials.

Table 9: User Stories

7.2 Critical Use Cases

Use Case 1: Borrow a Library Item

Preconditions:

- Member is logged in.
- Book is available in the catalog.

Main Flow:

1. Member presents book to Librarian.
2. Librarian scans member ID and Book Barcode.
3. System links book to member and calculates due date.
4. System updates book status to "Borrowed".

Alternate Flow:

- If member has unpaid fines exceeding limit, system blocks checkout.

Postconditions:

- Book is unavailable in catalog.
- Loan appears on Member's dashboard.

Use Case 2: Query AI Agent About Research Paper

Preconditions:

- User is authenticated or using public access.
- Research paper is accessible to user.
- AI Agent is operational.

Main Flow:

1. User opens AI chat.
2. User submits query about a paper.
3. System processes query and identifies document.
4. AI Agent retrieves content and generates response.
5. System displays answer with citations.

Alternate Flow:

- If access restricted, system denies content and suggests alternatives.

Postconditions:

- Query is logged.

Use Case 3: Get AI-Powered Book Recommendations

Preconditions:

- User is authenticated.
- Borrowing history exists.

Main Flow:

1. User requests recommendations.
2. AI analyzes history.
3. System retrieves relevant books.
4. Results ranked and displayed.

Postconditions:

- Recommendations shown with availability.

Use Case 4: Pay Library Fine

Preconditions:

- Member is logged in.
- Outstanding fine exists.

Main Flow:

1. Member views outstanding fines.
2. Member selects payment method.
3. System initiates payment transaction.
4. Payment gateway processes payment.
5. System confirms payment and updates balance.

Alternate Flow:

- If payment fails, system notifies user and allows retry.

Postconditions:

- Fine balance updated.
- Transaction recorded.

Use Case 5: Register/Login via SSO

Preconditions:

- User has institutional credentials.

Main Flow:

1. User selects "Login via Institution".
2. System redirects to SSO provider.
3. User authenticates externally.
4. System receives authentication token.
5. User session is created.

Alternate Flow:

- If authentication fails, access is denied.

Postconditions:

- User is logged in with appropriate role.

Use Case 6: External Researcher Access

Preconditions:

- User is not affiliated with institution.

Main Flow:

1. User registers as external researcher.
2. System verifies email and identity.
3. System assigns limited access role.
4. User browses public/restricted content.

Alternate Flow:

- Access to restricted materials is denied.

Postconditions:

- External account created with limited permissions.

Use Case 7: Upload Research with Review Workflow

Preconditions:

- User is authenticated (researcher/staff).

Main Flow:

1. User uploads research document.
2. User enters metadata.
3. System stores document in pending state.
4. Reviewer evaluates submission.
5. Reviewer approves or rejects document.
6. System updates publication status.

Postconditions:

- Approved content becomes searchable.

Use Case 8: AI Query with Access Restriction Failure

Preconditions:

- User submits AI query.
- Requested content is restricted.

Main Flow:

1. AI receives query.
2. System checks access permissions.
3. Restricted content is blocked.
4. AI generates alternative response using accessible sources.

Postconditions:

- No restricted data is exposed.

Use Case 9: Handle Payment Failure

Preconditions:

- Payment process initiated.

Main Flow:

1. Payment attempt fails.
2. System receives failure response.
3. User is notified with error message.
4. User retries or selects different payment method.

Postconditions:

- No incorrect charge recorded.

8 Data Requirements

8.1 Core Entities

Entity	Description	Key Fields	Constraints
User	System actors (Student, Staff, Admin)	UserID, Name, Email, RoleTier, CreatedDate	Email must be unique; Role is required.
Media / Book	Physical or digital item	ItemID, Title, Format, Status, CreatedBy, UploadDate	Status must be valid (Available, Borrowed, Draft, Archived).
Loan	Borrowing transaction	LoanID, UserID, ItemID, CheckoutDate, DueDate, ReturnDate	Cannot checkout an already borrowed ItemID. ReturnDate is nullable until return.
Metadata	Search tags and details	MetaID, ItemID, Tags, Abstract, Keywords, Language	Linked 1:1 with Media/-Book entity.
Research Output	Research papers and publications	ResearchID, Title, Authors, Abstract, UploadDate, Status	Status: Draft, Review, Published. Linked to Authors (N:M).
Student Project	Student coursework and projects	ProjectID, Title, TeamMembers, Supervisor, Description, Year, SubmissionDate	Status: Draft, Approved, Published. Academic year is required.
Payment	Records all fine payment transactions made by users	PaymentID, UserID, FineID, Amount, PaymentMethod, PaymentStatus, Timestamp	Amount > 0; PaymentStatus $\in$ {Pending, Successful, Failed}; Each payment must be linked to a valid Fine; Timestamp is system-generated.
AI Chat Message	AI Agent conversation record	**MessageID, UserID, SessionID, UserQuery, AIResponse, Timestamp**	**Linked to User and content documents; timestamp for session tracking.**
Vector Embedding	Semantic representation of content	**EmbeddingID, ItemID, EmbeddingVector**	**Linked 1:1 with content items; indexed for similarity search**

8.2 Data Dictionary

Field	Type	Required	Validation / Constraint	Example
DueDate	DateTime	Yes	Must be strictly > CheckoutDate	`2026-03-15 12:00:00`
RoleTier	Enum	Yes	Public, Member, Staff, Admin	`Staff`
Barcode	String	Yes	Must be unique, alphanumeric	`CSEDU-LIB-9932`
EmbeddingVector	Float Array	Yes	Dimension depends on model (typically 768-1536)	[0.123, -0.456, ..., 0.789]
UserQuery	Text	Yes	Length $\leq$ 5000 characters; supports multilingual Unicode	"What are the main findings of this paper?"
AIResponse	Text	Yes	Length variable; includes source citations	"Based on [Paper ID], the findings include..."

Table 11: Data Dictionary

8.3 Data Model and Relationships

Core Relationships:

- A *User* can have many *Loans* (1 : N).
- A *User* can create many *Media/Research* items (1 : N).
- A *Media/Book* item has one set of *Metadata* (1 : 1).
- A *Loan* links one *User* to one *Media/Book* (1 : 1 per active transaction).
- A *Research Output* has multiple *Authors* (N : M).
- A *Student Project* has multiple *Team Members* (N : M).

AI-Specific Relationships:

- **Each *Media/Research/Project* item has one *Vector Embedding* (1:1).**
- **A *User* can have many *AI Chat Messages* (1:N).**
- **Each *AI Chat Message* can reference multiple *Source Documents* (N:M).**
- **All *AI Chat Messages* are logged to *AI Performance Log* (1:N).**

8.4 AI Knowledge Base Structure

The AI Agent leverages the following data structures:

1. **Document Store:** All platform content (research papers, project descriptions, catalog entries, archives) is stored with full text indexing and metadata.
2. **Vector Database:** Semantic embeddings generated from document content enable similarity-based retrieval. Vectors are computed using a pre-trained language model (e.g., BERT, Sentence-Transformers).
3. **Metadata Index:** Structured metadata (titles, authors, publication dates, access tiers) is indexed for faceted filtering and ranking.
4. **Access Control Layer:** User permissions are applied during retrieval to ensure AI responses only reference accessible content.
5. **Citation Mapping:** Document IDs and source references are maintained to enable citation generation in AI responses.
6. **Conversation Context:** User sessions maintain conversation history for multi-turn context awareness.

9 UI and UX Requirements

9.1 Wireframes and Layout

- **Public Home / Showcase Grid:** Search bar centered, featured student projects below, AI Agent chat button in bottom-right corner.
- **Catalog Search Results:** Filters on the left (Format, Language, Year), list of items on the right, with "Ask AI" button for each result.
- **Media Viewer / Item Detail:** Metadata listed, “Request Hold” button, integrated PDF/Video viewer, AI-generated summary section (if available).
- **Member Dashboard:** Tabs for Active Loans, History, Fines, and Wishlist. Personalized AI recommendations section.
- **Librarian POS / Dashboard:** Large Barcode input field, quick-action buttons for Checkout/Return, Overdue alerts, embedded AI Chat for patron support.
- **AI Agent Chat Widget:** Collapsible floating chat interface accessible from all pages. Shows conversation history and allows multi-turn interaction.
- **AI Admin Panel:** Dashboard displaying query logs, performance metrics, vector database status, and model configuration options.

9.2 UX and Accessibility Baseline

- Clear navigation and labels (Bilingual support for both platform UI and AI responses).
- Keyboard accessibility for core flows (Search, Tab navigation for Librarian rapid entry, chat interface).
- Readable contrast and typography adhering to WCAG 2.1 AA standards.
- Responsive behavior for major screens (Mobile-friendly catalog search, collapsible chat widget on small screens).
- **AI Response Clarity:** AI-generated responses use plain language, clear formatting with bullet points, and include source citations.
- **Loading Indicators:** Users are informed when AI is processing queries (asynchronous response expectations communicated).
- **Error Handling:** Clear error messages when AI cannot answer queries; suggestions for refining questions.

9.3 AI Agent Interface Design

- **Chat Widget Placement:** Bottom-right corner of screen, persistent across all pages, minimizable but easily accessible.
- **Message Format:** User messages aligned to right, AI responses aligned to left, with source document links embedded.
- **Response Actions:** Users can rate responses (thumbs up/down), copy response text, or cite in papers.
- **Suggested Queries:** AI offers follow-up suggestions or related queries to guide users.
- **Content Preview:** Hovering over cited documents shows preview cards with title, author, and snippet.
- **Language Toggle:** Explicit language selector in chat widget for English/Bangla.

10 Non-Functional Requirements

NFR ID format: `NFR-TXX-###`

10.1 Performance Requirements

NFR ID	Category	Requirement	Target / Metric
NFR-TXX-001	Performance	Catalog search queries must return results quickly.	< 2 seconds for indexed catalog queries.
NFR-TXX-002	Performance	Full-text search response time for archive and research content.	< 2 seconds for > 90% of queries.
NFR-TXX-003	Performance	AI semantic search must retrieve relevant results efficiently.	< 5 seconds for vector similarity search on 100K+ documents.
NFR-TXX-004	Performance	AI response generation including retrieval and synthesis.	< 10 seconds for complete AI Agent response.
NFR-TXX-005	Performance	Document summarization response time.	< 10 seconds for documents up to 5 pages.
NFR-TXX-006	Performance	Page load time for public-facing interfaces.	< 3 seconds for 95th percentile of users.
NFR-TXX-041	Performance	File upload performance.	• Files ≤ 20MB upload within 5 seconds (90% cases) • Upload progress feedback visible to user

Table 12: Performance Requirements

10.2 Capacity and Scalability Requirements

NFR ID	Category	Requirement	Target / Metric
NFR-TXX-007	Capacity	System must support media file uploads.	• Video ≤ 200MB • PDF ≤ 100MB • Image ≤ 10MB • Audio ≤ 50MB
NFR-TXX-008	Capacity	Total archive document capacity.	Support 100,000+ documents with searchable indexes.
NFR-TXX-009	Capacity	Vector database scaling for semantic search.	Support 100,000+ embeddings with < 1 second retrieval.
NFR-TXX-010	Capacity	Concurrent user support.	• Support 500+ concurrent users • Response time degradation ≤ 20% under peak load
NFR-TXX-011	Scalability	Database scaling for growth.	Horizontal scaling supported via partitioning and indexing.
NFR-TXX-012	Scalability	AI Agent concurrent query handling.	Process 100+ simultaneous AI queries without failure.
NFR-TXX-013	Scalability	Document ingestion throughput.	Process 1,000+ documents per day for embedding generation.

Table 13: Capacity and Scalability Requirements

10.3 Security and Access Control Requirements

NFR-TXX-014 • Unauthorized access attempts blocked • Access violations logged • Access checks enforced on each request	Security	Role-based access control (RBAC) enforcement.
NFR-TXX-015 • JWT-based authentication • Session timeout: 30 minutes inactivity • Password hashing using bcrypt • Token invalidation on logout	Security	Authentication and session management.
NFR-TXX-039 • OAuth 2.0 / SAML supported • Access token expiry $\leq$ 1 hour • Refresh tokens supported • Max 5 failed login attempts per minute	Security	Single Sign-On (SSO) support.
NFR-TXX-016 HTTPS/TLS for all communication; AES-256 for stored data.	Security	Data encryption for sensitive information.
NFR-TXX-017 All CRUD operations logged; retention $\geq$ 90 days.	Security	Audit logging of system activity.

NFR ID Target / Metric	Category	Requirement
NFR-TXX-018 Restricted content filtered before response generation.	Security	AI Agent access control enforcement.
NFR-TXX-019 • $\geq$ 95% responses include source citations • Unsupported claims avoided or flagged • Low-confidence queries trigger fallback	Security	AI response reliability and citatio
NFR-TXX-020 Quarterly audits; zero critical vulnerabilities allowed.	Security	Vulnerability testing and security audits.

10.4 Reliability and Availability Requirements

NFR ID	Category	Requirement	Target / Metric
NFR-TXX-021	Reliability	System uptime and availability.	99% uptime; max 3.6 hours downtime/month.
NFR-TXX-022	Reliability	Backup and recovery.	Daily backups; 30-day retention; recovery tested quarterly.
NFR-TXX-023	Reliability	Disaster recovery.	RTO: 4 hours; RPO: 1 hour.
NFR-TXX-024	Reliability	Data integrity.	ACID compliance; zero data corruption incidents.
NFR-TXX-025	Reliability	AI fallback mechanism.	Fallback to keyword search if AI unavailable.
NFR-TXX-040	Reliability	API reliability.	$\geq$ 99% successful request rate; timeout $\leq$ 10 seconds.

Table 15: Reliability and Availability Requirements

10.5 Usability and Accessibility Requirements

NFR ID	Category	Requirement	Target / Metric
NFR-TXX-026	Usability	Responsive UI design.	Fully functional across mobile, tablet, desktop.
NFR-TXX-027	Usability	Bilingual support.	All UI available in English and Bangla.
NFR-TXX-028	Usability	Search usability.	Faceted filters, sorting, autocomplete supported.
NFR-TXX-029	Usability	AI response clarity.	Structured output with bullets and citations.
NFR-TXX-030	Accessibility	WCAG compliance.	WCAG 2.1 Level AA compliance.
NFR-TXX-031	Accessibility	Screen reader support.	ARIA labels and semantic HTML used.
NFR-TXX-032	Usability	AI interaction feedback.	Loading indicators and progress shown.

Table 16: Usability and Accessibility Requirements

10.6 Quality and Correctness Requirements

NFR ID	Category	Requirement	Target / Metric
NFR-TXX-033	Quality	AI response correctness.	• $\geq$ 85% correct answers on benchmark queries • $\geq$ 90% responses include citations
NFR-TXX-034	Quality	Summarization quality.	• Includes key findings and conclusions • $\geq$ 80% alignment with source
NFR-TXX-035	Quality	Semantic search relevance.	NDCG@10 $>$ 0.80; precision $>$ 80%.
NFR-TXX-036	Quality	Keyword search relevance.	$>$ 85% precision for top results.
NFR-TXX-037	Quality	Metadata completeness.	$>$ 90% documents have required fields.
NFR-TXX-038	Quality	Multilingual parity.	Equivalent functionality in English and Bangla.

Table 17: Quality and Correctness Requirements

11 Risks and Open Questions

11.1 Risk Register

RISK-001	Legacy data migration failure or corruption during bulk import	Medium	High	Perform dry-run migrations on staging database; validate data completeness; maintain backups; incremental migration approach.
RISK-002	Storage costs for large video media exceeding budget	High	Medium	Implement video compression on upload; establish storage quotas per user; use tiered storage; monitor costs monthly.
RISK-003	Librarian resistance to barcode/QR checkout workflow	Low	High	Conduct hands-on training sessions 2-3 weeks before launch; provide dedicated support staff during transition; phased rollout; feedback collection.
RISK-004	Security breach or unauthorized data access	Medium	High	Implement strong authentication (2FA); regular security audits; penetration testing; encryption at rest and in transit; access logging.
RISK-005	AI Agent generating hallucinations or false information	Medium	High	Mandatory citation enforcement; quality review pipeline; user feedback loop; regular accuracy audits; LLM fine-tuning; hallucination detection.
RISK-006	Vector database performance degradation at scale	Low	High	Load testing before production; index optimization; monitoring alerts; scalable infrastructure; fallback to traditional search.

(Risk Register – continued)

Risk ID	Description	Probability	Impact	Mitigation Strategy
RISK-007	LLM API rate limiting or service interruptions	Medium	Medium	Cache frequently queried summaries; implement request queuing; fallback responses; multiple LLM provider options; SLA agreements.
RISK-008	Poor translation or language quality in Bangla	Low	Medium	Use native Bangla speakers for review; test with multilingual models; collect user feedback; iterate on model selection.
RISK-009	Barcode scanner hardware failures	Low	Medium	Maintain scanner redundancy; establish vendor support contracts; provide fallback manual entry; regular hardware maintenance.
RISK-010	Insufficient AI testing leading to poor user experience	Medium	High	Dedicated AI QA team; extensive user acceptance testing; external expert review; alpha/beta testing with users; quality metrics.
RISK-011	Scalability issues with 500+ concurrent users	Low	High	Load testing with target user count; database optimization; caching strategies; horizontal scaling architecture; monitoring.
RISK-012	Integration with institutional authentication system delays	Medium	Medium	Early coordination with IT; parallel development; mockup authentication for testing; contingency local auth system.

(Risk Register – continued)

Risk ID	Description	Probability	Impact	Mitigation Strategy
RISK-013	Insufficient bilingual search quality	Medium	Medium	Separate indexing for EN/BN; use bilingual search models; native speaker testing; continuous improvement process.
RISK-014	Document versioning creating data complexity	Low	Medium	Clear versioning scheme; automated cleanup of old versions; archival policy; comprehensive testing.
RISK-015	Change management and user adoption challenges	Medium	Medium	Change management plan; user communication campaign; training materials; phased rollout; feedback mechanisms; support team.

Table 18: Risk Register and Mitigation Strategies

11.2 Open Questions and Decisions Pending

1. **Cloud Storage Provider:** Which cloud storage provider will CSEDU use for hosting large media files (AWS S3, Google Cloud Storage, Azure Blob Storage, or on-premise)?
2. **Fine Calculation Rules:** What are the exact university rules for calculating library fines? Specific amounts per day, grace periods, maximum fines, and enforcement thresholds?
3. **LLM Selection:** Which LLM provider and model will be used? (OpenAI GPT-4, Anthropic Claude, open-source Llama, or Mistral?)
4. **Vector Database:** Which vector database will be deployed? (Pinecone, Weaviate, Milvus, or other?)
5. **Authentication System:** Will institutional SSO (SAML/OAuth) be available or must local authentication be maintained?
6. **Budget Constraints:** What is the annual budget for cloud services, LLM API costs, and infrastructure?
7. **Timeline Confirmation:** Is the MVP delivery timeline (project lab deadline) con-

firmed, or does it need revision?

8. **Bangla Content:** How much existing institutional content is in Bangla vs. English? What are language preferences by user group?
9. **Data Governance:** Are there policies on data retention, archival, and deletion for research papers and archives?
10. **AI Governance:** Should an AI ethics committee or policy framework be established for the AI Agent?

12 Testability and Traceability

12.1 Requirement Traceability Matrix

FR-TXX-001	US-006	Upload Research with Metadata	TC-001, TC-002, TC-003
FR-TXX-002	US-001, US-012	Bilingual Search	TC-004, TC-005, TC-006
FR-TXX-003	US-008, US-014	Borrow a Library Item	TC-007, TC-008, TC-009
FR-TXX-004	US-011	Role-Based Access Control	TC-010, TC-011, TC-012
FR-TXX-005	US-004	Fine Calculation	TC-013, TC-014, TC-015
FR-TXX-006	US-007	Pay Library Fine	TC-016, TC-017, TC-018
FR-TXX-007	US-009	Handle Payment Failure	TC-019, TC-020
FR-TXX-008	US-010	Register/Login via SSO	TC-021, TC-022, TC-023
FR-TXX-009	US-013	External Researcher Access	TC-024, TC-025
FR-TXX-010	US-015	Upload Research with Review Workflow	TC-026, TC-027, TC-028
FR-TXX-011	US-016	AI Query with Access Restriction Failure	TC-029, TC-030
FR-AI-001	US-AI-001, US-AI-002	Query AI Agent About Research Paper	TC-AI-001, TC-AI-002, TC-AI-003
FR-AI-002	US-AI-004, US-AI-010	Semantic Search	TC-AI-004, TC-AI-005, TC-AI-006
FR-AI-003	US-AI-012, US-AI-015	Get AI-Powered Book Recommendations	TC-AI-007, TC-AI-008, TC-AI-009

12.2 Comprehensive Manual Test Plan

TC-001	Upload Research PDF	Staff logged in; valid PDF < 50MB	1. Open Upload Page 2. Select PDF 3. Enter metadata 4. Submit	File stored; status = Draft; visible in staff dashboard; metadata indexed.
TC-002	Upload Video Content	Staff logged in; valid video file	1. Upload video 2. Enter metadata 3. Submit	Upload accepted; async processing starts; playable within defined SLA.
TC-003	Bulk Import Records	Librarian; valid CSV	1. Upload CSV 2. Map fields 3. Confirm import	Records inserted; invalid rows logged; summary report generated.
TC-004	English Search	Public user	1. Enter keyword 2. Submit search	Results returned < 2s; ranked by relevance.
TC-005	Bangla Search	Public user	1. Enter Bangla query 2. Submit	Results in Bangla; accuracy > 85%.
TC-006	Bilingual Search	Public user	1. Enter mixed query 2. Submit	Handles EN+BN terms; no errors; relevant results returned.
TC-007	Borrow Book	Librarian; valid member	1. Scan member ID 2. Scan book 3. Confirm	Loan created; due date assigned; status updated.
TC-008	Return Book	Book is borrowed	1. Scan book 2. Confirm return	Status = Available; fine calculated if overdue.
TC-009	Invalid Barcode	Librarian logged in	1. Scan invalid code	Error shown; retry option available.
TC-010	Unauthorized Access	Public user	1. Access restricted URL	Access denied (403); no data exposed.
TC-011	Role Access Check	Staff user	1. Access member-only content	Access granted; restricted for public users.
TC-012	Role Update	Admin modifies role	1. Change role 2. Re-login	New permissions applied immediately.

(Manual Test Plan – continued)

Test ID	Scenario	Preconditions	Test Steps	Expected Result
TC-013	Fine Calculation	Overdue item exists	1. System runs daily job	Fine calculated correctly (rate-based).
TC-014	Block Borrow on Fine	User has unpaid fine	1. Attempt checkout	Operation blocked; message displayed.
TC-015	Fine Payment	User pays fine	1. Record payment	Balance updated; audit log stored.
TC-016	SSO Login	Valid institutional account	1. Click SSO login 2. Authenticate	User logged in; session created securely.
TC-017	Payment Gateway Success	User has fine	1. Initiate payment 2. Complete transaction	Payment successful; receipt generated.
TC-018	Payment Failure Handling	Payment interrupted	1. Attempt payment 2. Cancel/Fail	System rolls back; no balance change.
TC-019	External Researcher Access	External user registered	1. Login 2. Search archive	Limited access enforced; restricted content hidden.
TC-020	Submission Workflow	Student logged in	1. Submit project	Status = Pending; not public.
TC-021	Approval Workflow	Staff reviewing	1. Approve submission	Status = Published; visible publicly.
TC-022	Responsive UI	Mobile device	1. Open system	Layout adapts; usable interface.
TC-023	Language Toggle	Any user	1. Switch language	UI updates fully to selected language.
TC-024	Report Generation	Librarian	1. Generate report	Report generated with correct data.
TC-025	Metadata Editing	Staff user	1. Edit metadata 2. Save	Changes persisted; search index updated.
TC-026	Dashboard Load	Librarian	1. Open dashboard	Loads < 3s; real-time data shown.

(Manual Test Plan – continued)

Test ID	Scenario	Preconditions	Test Steps	Expected Result
TC-AI-001	AI Summarization	Paper exists	1. Ask AI to summarize	Summary generated < 10s; key points retained.
TC-AI-002	Semantic Search	Archive populated	1. Enter natural query	Relevant results returned; precision $> 80\%$.
TC-AI-003	AI Citation	User query	1. Ask question	Response includes valid citations.
TC-AI-004	Bilingual AI	Bangla query	1. Ask in Bangla	Response in Bangla; coherent.
TC-AI-005	Multi-turn Chat	Active session	1. Ask follow-ups	Context preserved across turns.
TC-AI-006	AI Access Restriction	Restricted data exists	1. Ask about restricted doc	No restricted data exposed.
TC-AI-007	AI Recommendations	User profile exists	1. Ask for recommendations	Personalized results shown.
TC-AI-008	Book Recommendation	Borrow history exists	1. Ask AI	Relevant books suggested.
TC-AI-009	AI Monitoring	Admin	1. View dashboard	Metrics displayed correctly.

12.3 Automated Testing Strategy

Test Category	Framework/Tool	Scope	Acceptance Criteria
Unit Testing	Jest (JS), PyTest (Python)	Individual functions and methods	80%+ code coverage; all tests pass
Integration Testing	Postman, REST Assured	API endpoints and module interactions	All workflows functional; data consistency
UI Automation	Selenium, Cypress	Functional workflows	Core user journeys pass; cross-browser
Performance Testing	JMeter, Locust	Load testing at 500 concurrent users	Response times < 2s; memory stable
AI Quality Testing	Manual expert review	AI response accuracy and relevance	>85% accuracy; minimal hallucination
Security Testing	OWASP ZAP, Burp Suite	Vulnerability scanning; penetration test	Zero critical issues; zero auth bypasses

Table 21: Automated Testing Strategy